TITLE: ENDPOINT SECURITY INCIDENT RESPONSE AND AUTOMATED REMEDIATION ASSESSMENT REPORT

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EXECUTIVE SUMMARY

This report provides a secure endpoint monitoring lab based on Windows 11 VM and Kali Linux VM. The logs were sent to Splunk Enterprise, which saved the failed logins, privilege escalations, persistence methods and reconnaissance. The alerts were categorized and fixed automatically with the help of a Dockerized AI security stack. Remediation activities were listed, audit verifiable and were provided with verification and rollback procedures to make them safe.

The deployment shows end-to-end detection and response, which gives quantifiable gains in operational efficiency. The piece is consistent with the MITRE ATT&CK framework (MITRE, 2023) and NIST SP 800-61 incident handling advice (NIST, 2012), and makes use of the Splunk REST API to integrate with it safely (Splunk, 2025).

CONTEXT

- Name: Larkspur Retail Group.
- Sector: Retail, mixed office, warehouse and remote endpoint.
- Endpoint Instance: Windows 11 VM, Kali Linux VM
- Summary of the Incident: Some of the past attacks involved credential theft, suspicious scripts, and privilege escalations, and lateral movement.

AIM

Demonstrate a working lab which proves:

- Intentionally weak endpoints Secure lab build.
- Splunk SIEM Endpoint telemetry collection.
- Representative breaches detection engineering.
- Automated remediation through a Dockerized AI stack of security.

APPROACH

- Virtual machines and internal network built and isolated lab.
- Simulated breaches by using configured weak Windows and Linux endpoints.
- Splunk forwarders installed on both terminals.
- Analysis of attack signals (generated attack signal) (4625, 4732, 4698, sudo failures, nmap).
- Designed queries, and alerts in Splunk.
- Developed a Dockerised AI automation stack to categorize alerts and implement allowed remediation.

KEY FINDINGS

- Brutality detected with credentials (Windows 4625).
- Escalation of privilege succeeded (Windows 4732).
- Creation of scheduled tasks identified (Windows 4698).
- Reconnaissance on sudo failed, and nmap reconnaissance taken on Kali.
- Automated remediation that is properly executed is able to execute allowlisted actions (disable user, kill nmap process).
- End-to-end visibility, detection and remediation were lab confirmed.

KEY DELIVERABLES

- Splunk ingestion pipeline used on both windows and Linux endpoints.
- Authentication anomaly, process execution and network activity dashboards.
- Dockerized neural network scripting healing measures.
- Simulation scripts of benign breaches.
- Presentation of video on end-to-end workflow.
- Impact and Value Less likelihood of privilege exploitation and reconnaissance going undetected.
- Better operational efficiency through automated response.
- Mechanized system of tracking and correcting endpoint threats.

RECOMMENDATIONS

- Secure Windows and Linux systems with new OS and password policies.
- Minimum privileges on administrative accounts.
- The telemetry coverage should be expanded to cover network flow logs and file integrity monitoring.
- When false positives are too many, apply regular SIEM health checkups and alert tuning.
- Conducted regularly, automation playbooks are tested in a non-production setting.

TABLE OF CONTENTS

INTRODUCTION……..5

LITERATURE REVIEW AND STANDARDS ALIGNMENT…….……5

SCENARIO AND THREAT MODEL COMPANY…….……5

METHODOLOGY……..……..6

ARCHITECTURE AND IMPLEMENTATION OF THE SYSTEM……7

RESULTS……8

DISCUSSION……8

RECOMMENDATIONS……8-9

CONCLUSION……9

REFERENCES……9

APPENDICES……..10-15

VIDEO LINKS……..16

INTRODUCTION

Background

The Larkspur Retail Group has a low visibility mixed endpoint estate. Risk mitigation and efficient containment of breach should be conducted through automated detection and remediation.

Objectives and Scope

One Windows 11 VM, one Kali Linux VM, a Splunk SIEM, and a Dockerized AI automation stack were considered in scope. They included endpoint monitoring, detection, and automated remediation.

Out of Scope

Complete redesign of production network, email security as well as investigation of forensically beyond lab endpoints.

Report Structure

The report discusses design, laboratory configuration, breach testing, SIEM telemetry, automated remediation and suggestions.

LITERATURE REVIEW AND STANDARDS ALIGNMENT

The principles of endpoint security are based on least privilege, reducing the attack surface, and secure logging (NIST, 2020; CIS, 2021). Detection engineering is based on organized telemetry (Windows auditing, Splunk Forwarder, and Linux auditd) according to adversary techniques on the MITRE ATT&CK framework (MITRE, 2023). SIEM architectures offer a centralized ingestion, normalization and correlation of logs as well as their retention to facilitate meaningful monitoring and investigation (NIST, 2012). Predefined and allowlisted playbook response is a best practice of incident handling but must be governed to avoid false positives (NIST, 2018).

SCENARIO AND THREAT MODEL COMPANY

- Overview Retail organization mixed endpoints; sensitive customer data; file services associated with lateral movement.
- Simulated attacks were brute force, privilege escalation, and reconnaissance.
- Experienced attacker on traditional techniques (MITRE ATT&CK TTPs) in a laboratory environment.
- Assets and Trust Boundaries Endpoints, SIEM, network segments, and sensitive logs are considered as high-value telemetry.
- Passwords of poor quality, sudoers settings, administrative privileges uncovered.

METHODOLOGY

Lab Environment Build VMs used in the VirtualBox, isolated NAT network, IP addresses:

WINDOWS 11: 192.168.56.104

KALI LINUX: 10.0.3.15

UBUNTU (SPLUNK SIEM): 192.168.56.110

VMs were installed within the VirtualBox using an isolated NAT network. Windows 11 and Ubuntu (Splunk) are on the subnet of 192.168.56.x and Kali Linux is on the subnet of 10.0.3.x. The routing and port forwarding policies were set in such a way that telemetry traffic of Kali Linux can be forwarded to Splunk on Ubuntu without exposing any service to outside networks. This makes sure that there is centralization of all endpoint logs in Splunk without lab isolation.

Weak Security Configurations

Weak local passwords, administrative privileges, default logging

Windows: Weak local passwords, administrative privileges, default logging

Kali: misconfigurations, open test services, default accounts.

Telemetry Sources

Windows: Event log (4625, 4732, 4698), PowerShell.

Linux: Nmap execution logs, process accounting, auth.log.

SIEM Integration

Splunk Universal Forwarders in place. Indexes were made, normalization and log parsing.

The Detection Engineering Approach considers the risk factors and recognizes them based on the evidence available. MITRE ATT&CK mapping, event correlation, severity model. Brute force queries, privilege escalation queries, persistence queries, reconnaissance queries.

Remediation Design and Automation

Dockerized AI stack Polling Splunk to give alerts to and classify and run allowlisted remediation: kill nmap, disable testuser. Checking of actions through follow up inquiries.

Validation and Evaluation Tests performed

Windows login failure, user addition, nmap scan Kali sudo failure, and windows login failure. The logs were verified in Splunk dashboards. Confirmation of simulated success in remediation was done through automation logs.

Ethical and Safety Concerns

No live malware isolated, just simulations.

ARCHITECTURE AND IMPLEMENTATION OF THE SYSTEM

High-Level Architecture

The architecture is composed of windows and Linux nodes that relay the telemetry using Splunk Universal Forwarders to Splunk Enterprise that will index and correlate all the telemetry centrally. A Dockerized AI security stack is linked to the Splunk using its REST API to access the alerts and implement a set of preset remediation workflows.

Diagram (Appendix A): Windows and Linux Endpoints ,Splunk Forwarders Splunk Enterprise Docker AI Security Stack.

Endpoint Configurations

Windows: Splunk Forwarder was set up, PowerShell logging, and a local administrator account was utilized to simulate attacks.

Linux: sudo setup, auditd is set up to do authentication and monitor privileges, process accounting is enabled and nmap has been installed to perform controlled reconnaissance testing.

SIEM Configuration

Normalized fields were identified in custom indexes in order to provide efficient search and correlation. Authentication Anomalies, privilege escalation, processes and network activity were developed. Brute force attempts, privilege escalation events, scheduled task creation, sudo failures and reconnaissance activity were all put in place as alert rules.

AI Security Stack in Docker

It was done with Docker Compose having defined volumes, independent networks, management of secrets, and restriction of resources. The least-privilege container execution, limited filesystem access, and highly controlled execution of a set of established remediation measures are security controls that promote the prevention of uncontrolled command execution.

RESULTS

- Windows 4625 (failed login) → a series of several events of failed logging in.
- Windows 4732(admin addition) - user -->administrators.
- Windows 4698→create scheduled task.
- Attempts of sudo by Kali resulted in authentication failure entries.
- Kali nmap scans → the execution of processes recorded.

The detection uses cases that were implemented were effectively tested on both endpoints. High-severity alerts were caused by the use of brute force activity (Windows Event ID 4625, T1110) and privilege escalation (Event ID 4732, T1069), and medium-severity alerts were caused by persistence through scheduled tasks (Event ID 4698, T1053). Reconnaissance was identified using auditd log data and auth.log data on Linux by scanning with nmap (T1595) and by trying to failing to log in with sudo (T1078). The alerts were generated in Splunk correctly, and it is possible to confirm the effective ATT &CK aligned cross-platform detection.

Automated Remediation

- nmap terminated on Kali
- Windows simulated user disabled
- Checking through Splunk logs

Performance and Reliability

- Live ingestion.
- Significantly, the alerts were processed successfully by Docker stack.
- Delays of the polling (within 1-2 seconds) are minimal.

Limitations

Lab-based, no real malware User disable not imposed in manufacturing.

DISCUSSION

Effective detections based on specific sources of logs and correlation.

Automation minimizes its response time yet; it should be selectively allowlisted.

RECOMMENDATIONS

The weak passwords and the exposure of services should be removed to reduce the attack surface, with the process of hardening endpoints according to the secure configuration guidance (CIS, 2021). One of the top-priority controls is to enforce least privilege on user accounts because it restricts lateral movement and misuse of privileges (NIST, 2020). Increased

scanning of the telemetry will enhance the process of identifying more of the adversary techniques in line with the MITRE ATT&CK framework (MITRE, 2023). It is advisable to automate the process of playbook testing to minimize the number of false positives and enhance the reliability of incident responses (NIST, 2012). SIEM should be reviewed and validated regularly to maintain constant log ingestion, the accuracy of alerts, and operational efficacy (NIST, 2018).

CONCLUSION

This evaluation was able to prove:

- Windows and Linux endpoints End to end lab.
- Splunk telemetry data collection.
- Discovery of indicative violations.
- Docker AI stack Automated remediation.
- Every goal was accomplished, and the lab could be used as a safe prototype of production implementation.

REFERENCES

NIST 2012, Computer Security Incident Handling Guide, SP 800-61 Rev.2, NIST, Gaithersburg.

MITRE (2023) MITRE ATT&CK® Framework. The MITRE Corporation.

CIS (2021) CIS Controls Version 8. Center for Internet Security.

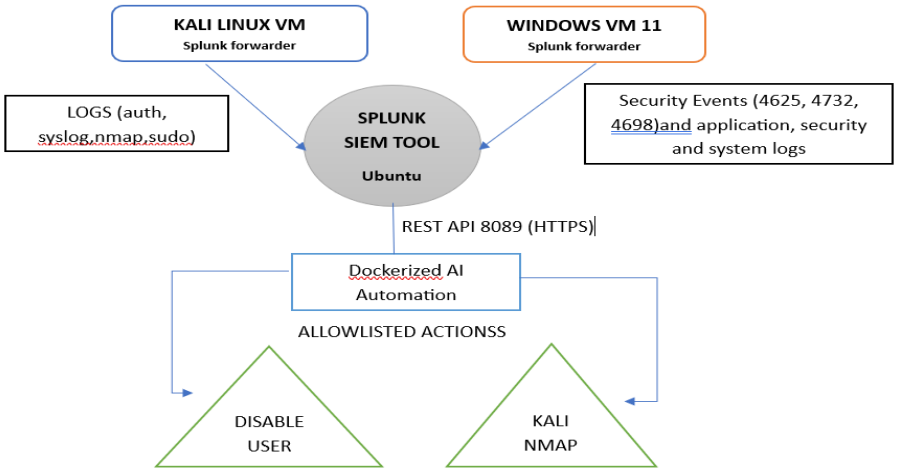
Splunk (2025) Splunk Enterprise REST API Reference Manual. Splunk Inc.

NIST (2020) Security and Privacy Controls for Information Systems and Organizations (SP 800-53 Rev. 5). Gaithersburg, MD: National Institute of Standards and Technology.

NIST (2018) Security Orchestration, Automation, and Response (SOAR) considerations (SP 800-184). Gaithersburg, MD: National Institute of Standards and Technology.

APPENDICES

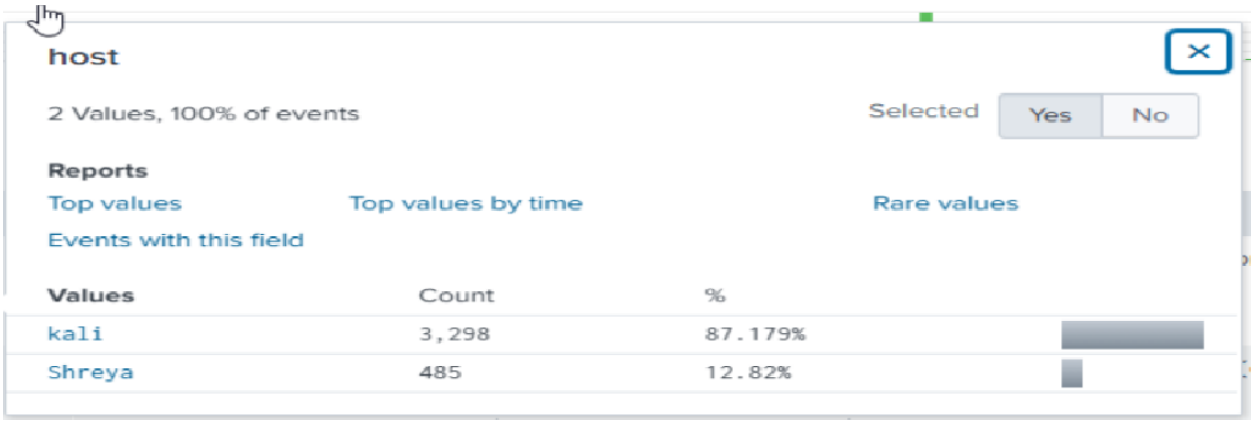
APPENDIX A - WINDOWS AND LINUX ENDPOINTS, SPLUNK FORWARDERS SPLUNK ENTERPRISE DOCKER AI SECURITY STACK.



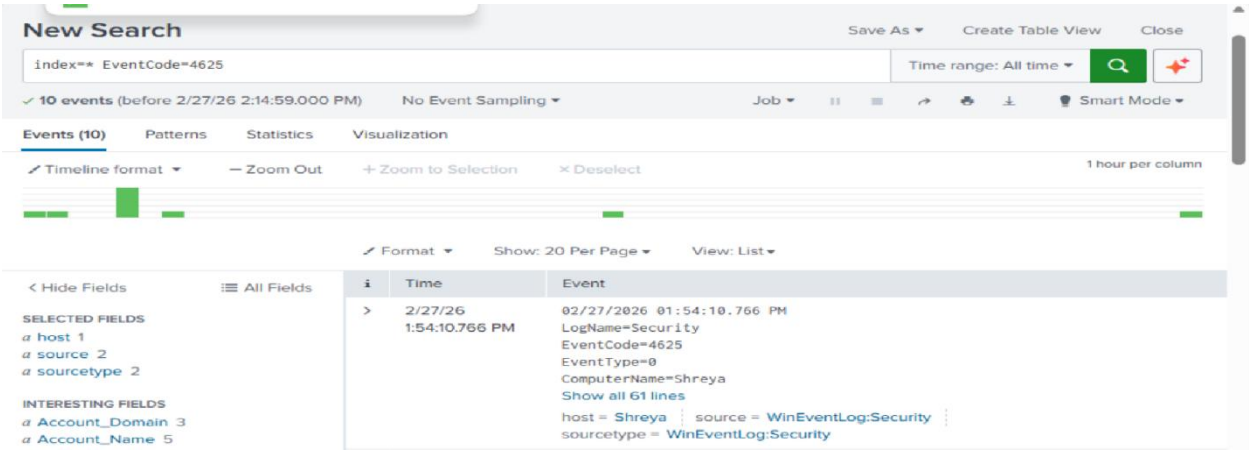
APPENDIX B – CREATION OF SPLUNK ENTERPRISE THROUGH UBUNTU SERVER

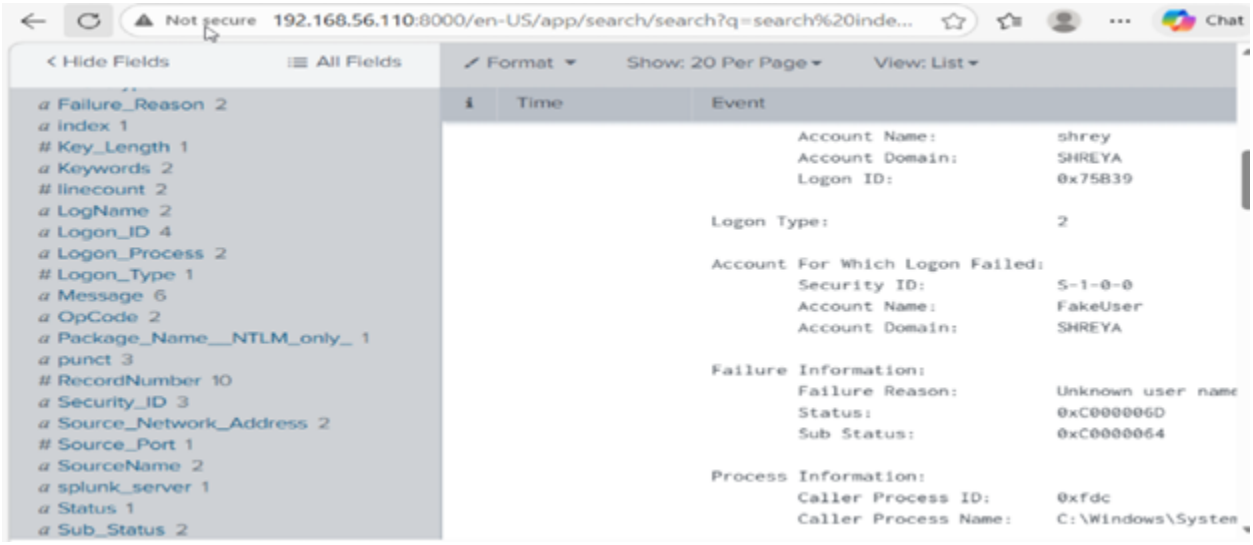
```
shreya@shreya:~$ docker run -d \
--name splunk \
--memory=4g \
-p 8000:8000 \
-p 8089:8089 \
-p 9997:9997 \
-v splunk-data:/opt/splunk/etc \
-e SPLUNK_START_ARGS="--accept-license" \
-e SPLUNK_PASSWORD="Shreya94" \
> splunk/splunk:latest
1ef27adad9ec6a64baa6b4e16fa59e75aad075116a13c49350663697a6a075a8
```

APPENDIX C – WINDOWS AND KALI LOGS WERE SENT TO SPLUNK ENTERPRISE

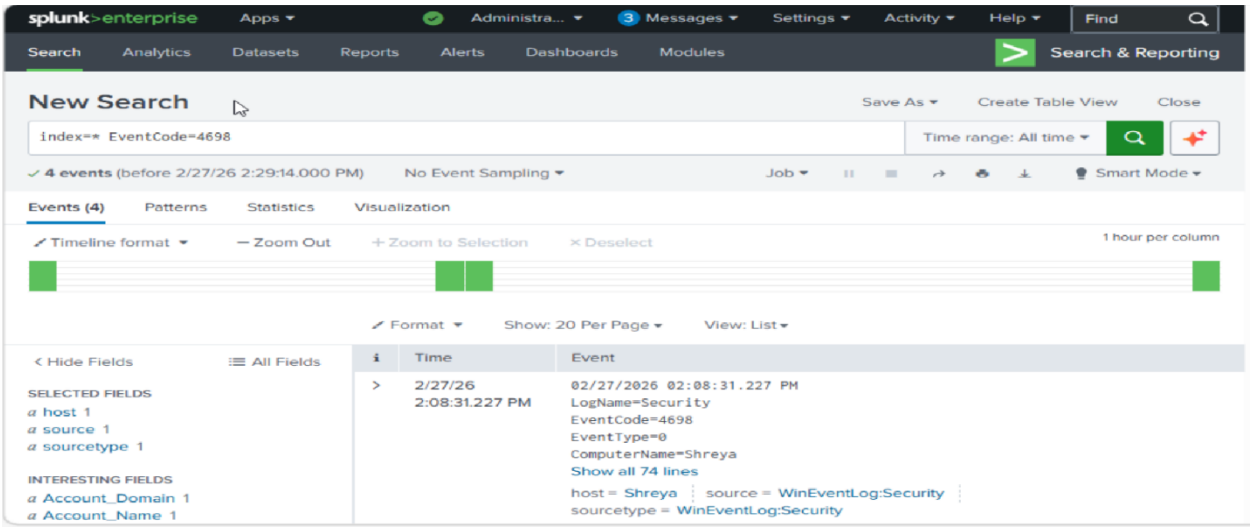


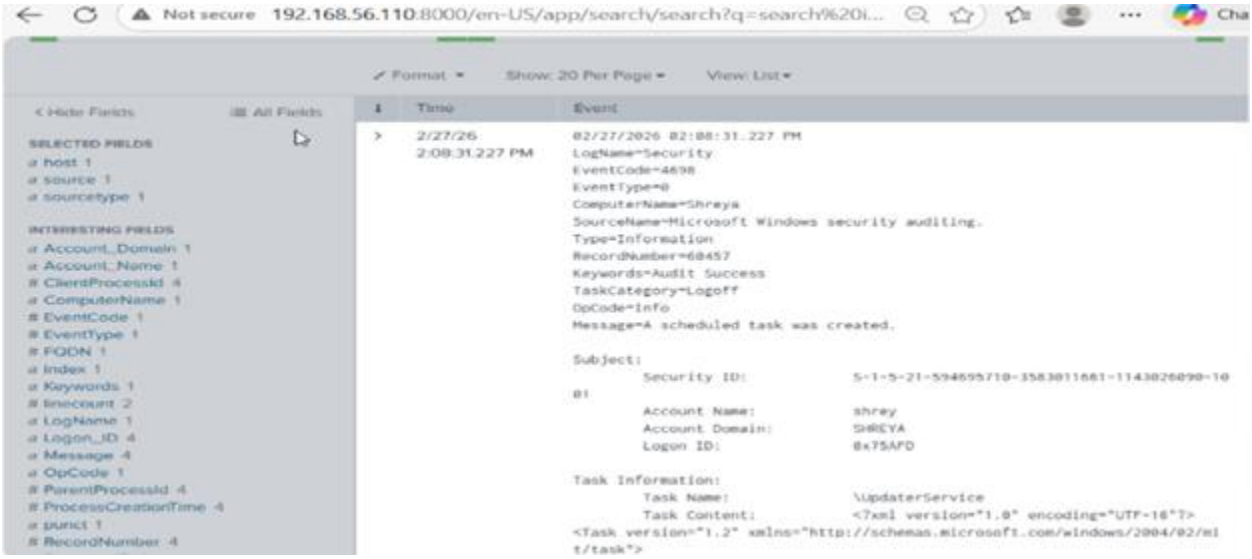
APPENDIX D – SECURITY EVENT 4625 WINDOWS



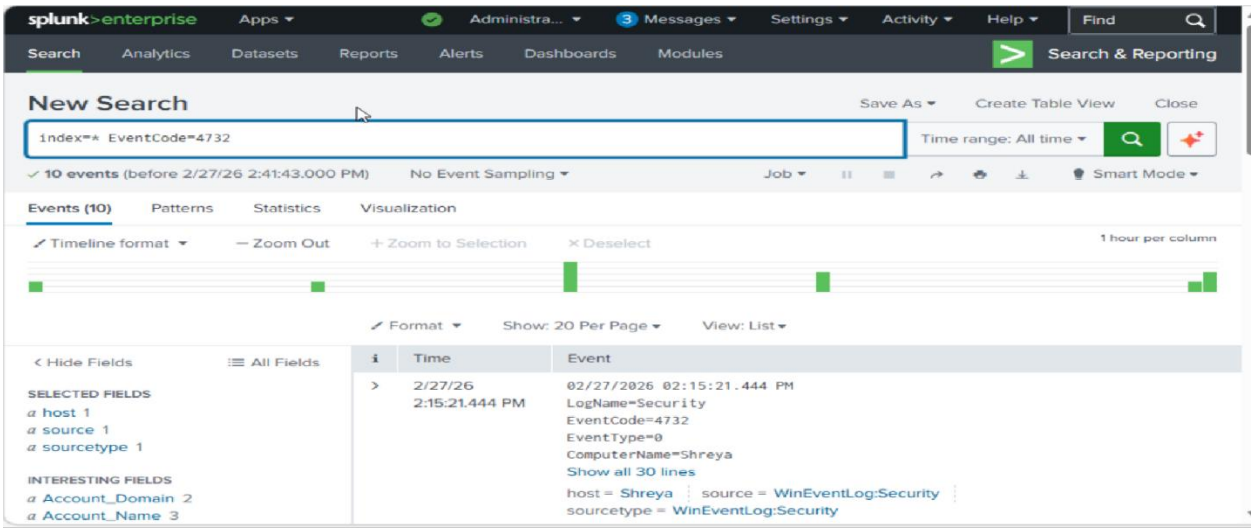


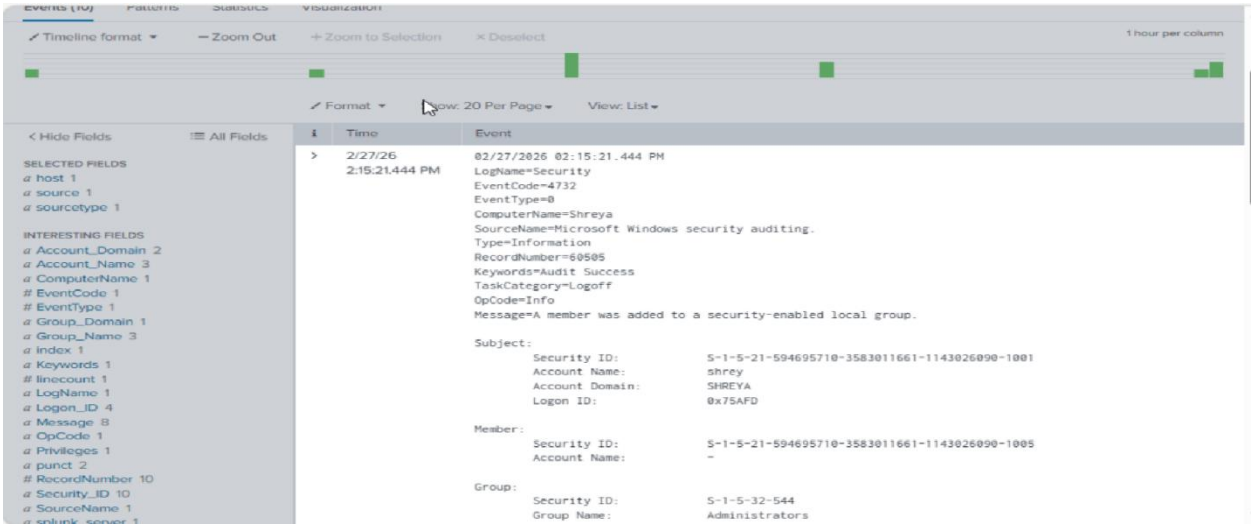
APPENDIX E – SECURITY EVENT 4698 WINDOWS



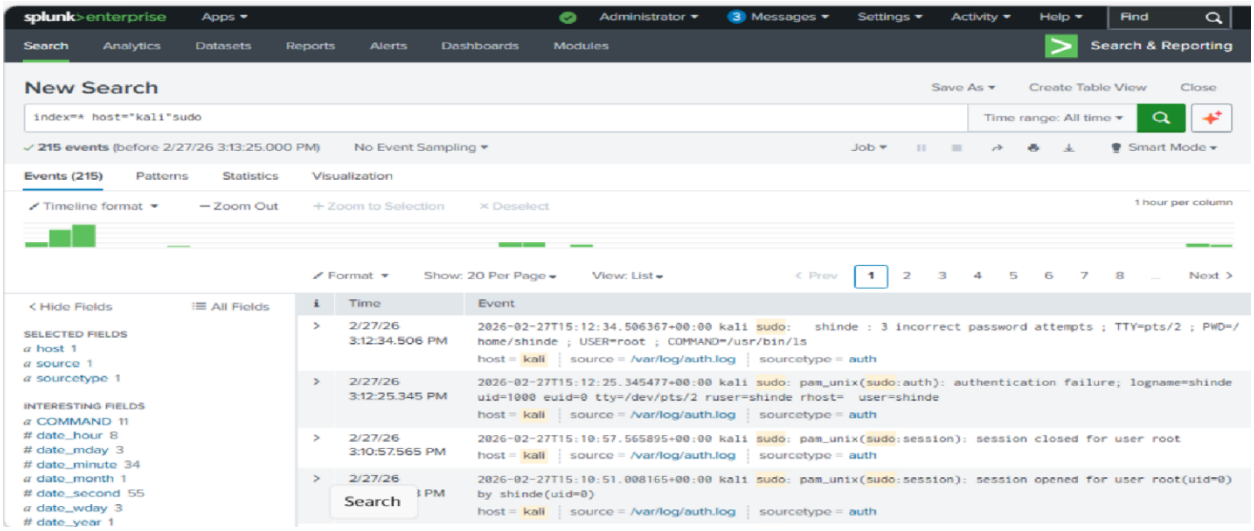


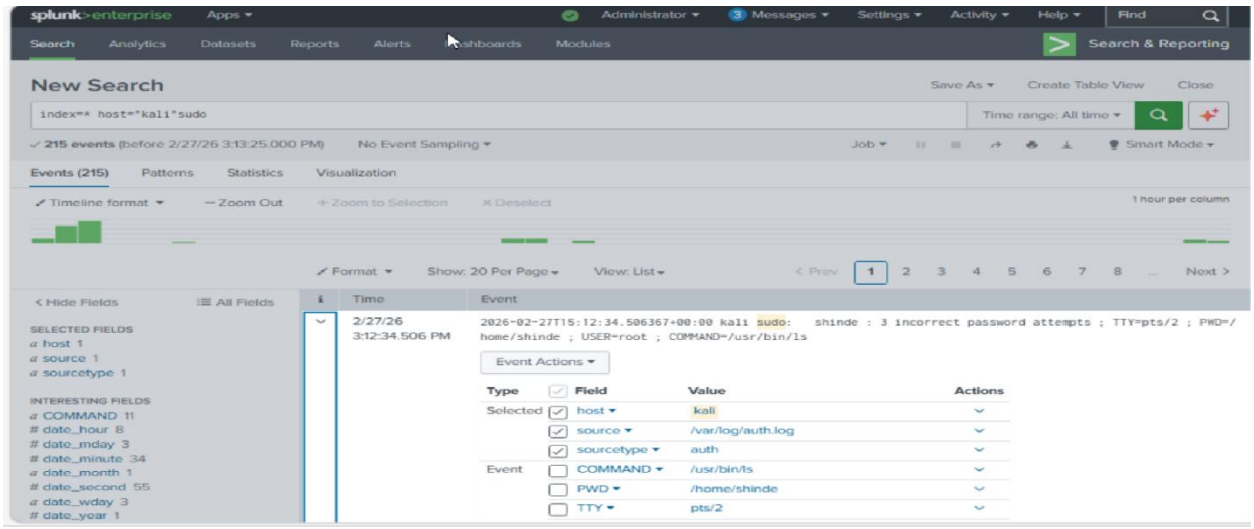
APPENDIX F – SECURITY EVENT 4732 WINDOWS



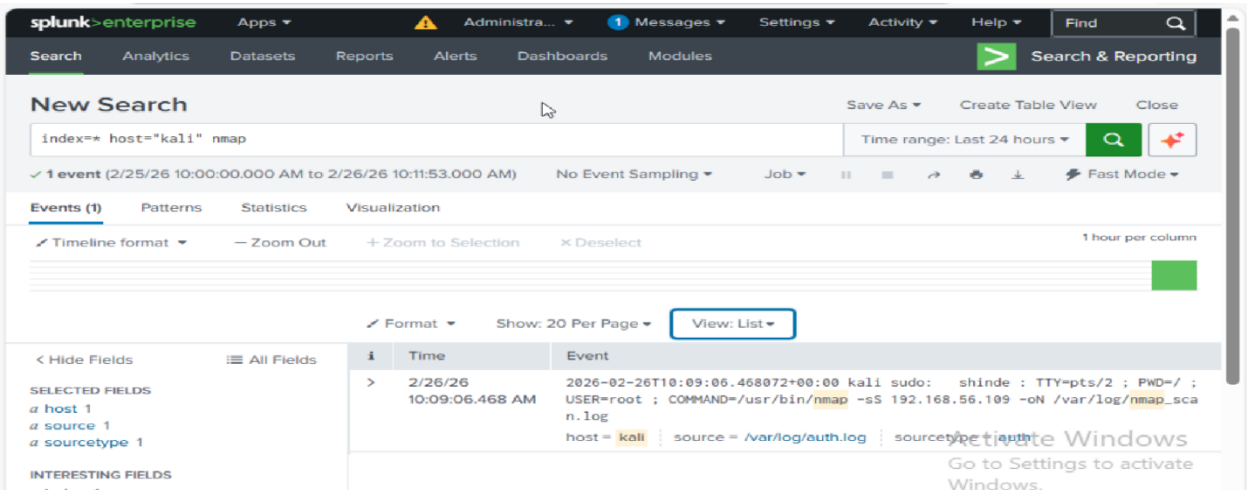


APPENDIX G – SUDO ATTEMPTS FAILURE KALI





APPENDIX H – NMAP SCANS KALI



APPENDIX I - USE CASES AND RESULTS TABLE

USE CASE	LOG SOURCE	MITRE ATTACK	TRIGGER CONDITION	RESULT
BRUTE FORCE LOGIN	WINDOWS SECURITY EVENT	T1110	EVENT CODE=4625	DETECTED, ALERT TRIGGERED
PRIVILEGE ESCALATION	WINDOWS SECURITY EVENT	T1069	EVENT CODE=4732	DETECTED, ALERT TRIGGERED
SCHEDULED TASK PERISISTENCE	WINDOWS SECURITY EVENT	T1053	EVENT CODE=T1053	DETECTED, ALERT TRIGGERED
SUDO FAILURE	LINUX auth.log	T1078	FAILED SUDO ATTEMPTS	DETECTED, ALERT TRIGGERED
RECONNAISSANCE	LINUX NMAP LOGS	T1595	EXECUTION OF NMAP SCAN	DETECTED, ALERT TRIGGERED

```
shreya@shreya:~/splunk_automation$ docker logs -f splunk_auto
[AI] Connected to Splunk
[AI] Checking for alerts...
[AI] Number of alerts fetched: 0
[AI] Checking for alerts...
[AI] Number of alerts fetched: 0
[AI] Checking for alerts...
[AI] Alert classified as: Recon Activity
[AI] Executing allowlisted action: kill_nmap
[AI] Verification successful: nmap terminated
[AI] Number of alerts fetched: 1
[AI] Checking for alerts...
[AI] Number of alerts fetched: 0
[AI] Checking for alerts...
[AI] Number of alerts fetched: 0
[AI] Checking for alerts...
[AI] Number of alerts fetched: 0
[AI] Checking for alerts...
[AI] Number of alerts fetched: 0
[AI] Checking for alerts...
[AI] Number of alerts fetched: 0
[AI] Checking for alerts...
[AI] Number of alerts fetched: 0
[AI] Checking for alerts...
[AI] Number of alerts fetched: 0
[AI] Checking for alerts...
[AI] Number of alerts fetched: 0
[AI] Checking for alerts...
[AI] Number of alerts fetched: 0
[AI] Checking for alerts...
[AI] Alert classified as: Brute Force Attempt
[AI] Executing allowlisted action: disable_user
Disabling Windows user testuser
[AI] User disable simulated for testuser
[AI] Verification successful: Windows user testuser disabled (simulated)
[AI] Alert classified as: Brute Force Attempt
[AI] Executing allowlisted action: disable_user
Disabling Windows user testuser
[AI] User disable simulated for testuser
[AI] Verification successful: Windows user testuser disabled (simulated)
[AI] Alert classified as: Brute Force Attempt
[AI] Executing allowlisted action: disable_user
Disabling Windows user testuser
[AI] User disable simulated for testuser
[AI] Verification successful: Windows user testuser disabled (simulated)
[AI] Number of alerts fetched: 3
[AI] Checking for alerts...
```

VIDEO LINKS:

https://www.loom.com/share/a45973f427394efe8b051d212fe19832

https://www.loom.com/share/862aecb90cfa4837a96798627725115f

https://www.loom.com/share/a04bd996398141b8b62857502f893494